

[Outdoor](#)[Husky](#)[Husky A300](#)[Integration](#)

Husky A300 Integration



CAUTION

Always perform a risk assessment prior to any custom integrations.

Custom integrations are outside the scope of the base robot's safety assessment and custom integrations may result in new hazards.

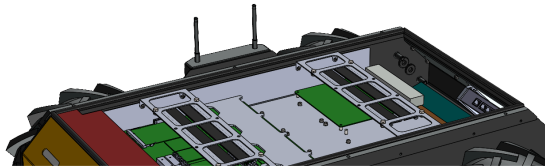
To attach custom hardware to Husky, you will have to take care of mechanical mounting, electrical supply, and software integration. This guide aims to equip you with respect to these challenges.

1. Mechanical Mounting

1.1 STEP Model

[Download STEP Model Of The Robot](#)

This simplified 3D model of the Husky A300 includes details like the Top Plate mounting surface, internal Battery Integration Volume, and the System Interface circuit board where you can connect to the robot's electrical provisions.



40 Ah battery configuration is shown. Other configurations are available.

1.2 base_link

`base_link` is the primary coordinate system for the robot. All sensors, manipulators, and brackets are positioned with reference to `base link`. [This ROS 2 Documentation page](#) explains how a robot's model is constructed.

Husky's model is included on the robot, and is available on our public GitHub. Husky's `base_link` is located on the bottom of the chassis, with +X pointed toward the front panel, and +Y toward the robot's left side.

! INFO

The [STEP Model](#)'s origin is aligned with `base_link`.

1.3 PACS™ Mounting and Kits

Husky A300 is equipped with a [PACS™](#) Top Plate for mounting sensors and equipment. [PACS™](#) is a Clearpath Robotics standard, providing a grid of M5×0.8 holes onto the Top Plate of the robot. This

grid of holes has a 80 mm X 80 mm spacing. You can create your own brackets to interface with these holes, or can use an existing Clearpath Robotics bracket.

NOTE

Our [Sensors](#) and [Add-ons](#) pages indicate the required bracket for the particular attachment.

1.4 Cable Passthrough

To maintain Husky's IP54 rating while allowing cables to pass between the external kits and internal electronics, Husky's top plate is provisioned with expansion plates for cable IP rated cable systems. There are three expansion plates at the front, and three at the rear of the top plate.

Top plate electrical expansion

Technicians have two main options.

1. Drill holes in the expansion plates and add appropriate IP-rated sealing solutions on a cable-by-cable basis.
2. Remove an expansion plate and replace it with a cable entry system, such as Icotek.

Options, from left to right: hole with a connector, Icotek straight housing, Icotek 90° housing

1.4.1 Using the Icotek Cable Entry System

The [Icotek Cable Entry System](#) allows pre-terminated cables up to 65 mm in diameter to be routed into Husky A300, and be sealed with up to IP66 rated ingress protection. In addition, the cable entry frames serve as strain relief on the cable.

This cable system is based on a set of cable entry frames and grommets. Refer to the [demonstration video](#) for an overview of using the system.

► Supported cable entry frames:

▶ Supported grommets:

1.5 Internal Integrations

There are interfaces inside the Husky for your custom components. This [Interface Control Drawing](#) shows the available volume near the robot's batteries, noting that the size of this volume changes depending on what battery configuration your Husky has—(40 Ah, 80 Ah, or 120 Ah). The drawing also shows mounting provisions on the robot's Electronics Tray, for including these components:

- Husky's System Interface Assembly (*circuit board with Husky's microcontroller, power supplies, and motor controllers*)
- Primary computer
- IMU
- Add-on secondary computer or graphics card
- Add-on USB hub

Refer to the [STEP model](#) for more measurements of the Husky's interface areas.

i NOTE

This integration volume is available inside the base Husky A300, as well as Husky AMP and Husky Observer.

1.6 Custom Robot Panel Colour

Your robot's yellow panels can be swapped for a custom colour. We use TIGER Drylac powder coating on these parts. TIGER has a wide range of colours, though some are special order.

Talk to a Clearpath Robotics salesperson to discuss custom colour prices and lead times: research-sales@clearpathrobotics.com

2. Electrical Integration

Refer to [Cable Passthrough](#) for details on how to pass cables through Husky's top plate for external payloads. |

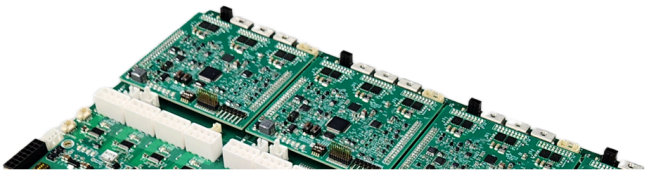
2.1 Data Connections

i NOTE

These are our suggested methods for connecting sensors and actuators to Husky.

Data Bus	Clearpath's suggestion for connecting devices to the Husky's computer:
USB	Direct connection to the robot's computer, or add a USB hub.
Ethernet	Connect to the robot's 2.5 Gbit/s Ethernet Switch, which is inside the chassis.
Serial RS-232, RS-422, RS-485	Add a USB converter, such as a FTDI cable. Most computers have Serial ports, but these take more time to configure than a FTDI cable.
CAN	Add a USB converter, such as a PEAK System PCAN-USB.

2.2 System Interface Connector Summary



The System Interface Assembly is a circuit board that includes power supplies, lighting controllers, motor controllers, and the Husky's microcontroller. This circuit board has connectors for: power, fans, relays, GPIOs, auxiliary inputs, auxiliary outputs, and Emergency Stop devices.

Purpose	Label on the circuit board	Connector on the circuit board	Mating connector	Mating connector's crimp	Crimping tool
User Power Connections	USER 1 - USER 8	Molex 0039296088	Molex 0039012080	Molex 0039000140 (AWG 22-28) Or Molex 0039000073 (AWG 18-24)	Molex 0638191000

Purpose	Label on the circuit board	Connector on the circuit board	Mating connector	Mating connector's crimp	Crimping tool
Expansion Power	EXP 1 - EXP 2	Molex 1720650202	Molex 1700010102	Molex 1720630311 (AWG 14-16) Or Molex 1720630312 (AWG 12)	Molex 2238631200
AUX Inputs	AUX1 IN, AUX2 IN, AUX3 IN	TE 3-794682- 6	TE 794617-6	TE 1-794606- 1 (AWG 20- 24) Or TE 1-794607- 1 (AWG 26- 30)	TE 91391-1
AUX Outputs	AUX1 OUT, AUX2 OUT, AUX3 OUT	TE 3-794682- 8	TE 794617-8	TE 1-794606- 1 (AWG 20- 24) Or TE 1-794607- 1 (AWG 26- 30)	TE 91391-1
GPIO	GPIO	TE 4-794680- 6	TE 1- 794617-6	TE 1-794606- 1 (AWG 20- 24) Or TE 1-794607- 1 (AWG 26- 30)	TE 91391-1

Purpose	Label on the circuit board	Connector on the circuit board	Mating connector	Mating connector's crimp	Crimping tool
Fans	FAN 5 - FAN 8	Molex 0705430003	Molex 0050579404	Molex 0016020103 (AWG 22-24)	Molex 0638118700
Wireless Emergency Stop	W ES (Wireless E-Stop)	TE 4-794682- 0	TE 1- 794617-0	TE 1-794606- 1 (AWG 20- 24) Or TE 1-794607- 1 (AWG 26- 30)	TE 91391-1
Emergency Stop Buttons	ES 3 - ES 8 (E-Stop)	TE 3-794682- 4	TE 794617-4	TE 1-794606- 1 (AWG 20- 24) Or TE 1-794607- 1 (AWG 26- 30)	TE 91391-1
Computer Power	SYS1 - SYS 2	Molex 1720650204	Molex 1700010104	Molex 1720630311 (AWG 14-16) Or Molex 1720630312 (AWG 12)	Molex 2238631200
Graphics Card Power	SYS3	Molex 1720650206	Molex 1700010106	Molex 1720630311 (AWG 14-16) Or	Molex 2238631200

Purpose	Label on the circuit board	Connector on the circuit board	Mating connector	Mating connector's crimp	Crimping tool
				Molex 1720630312 (AWG 12)	
Network Switch Power	SYS4	Molex 0039281023	Molex 0039012020	Molex 0039000140 (AWG 22-28) Or Molex 0039000073 (AWG 18-24)	Molex 0638191000

2.3 Power Connections

2.3.1 User Power Connections

The System Interface circuit board has 8 User Power connectors, labelled USER 1 through USER 8.

User power is divided into 4 power rails and the power for each rail is shared across each of the 8 connectors. The maximum—(*aggregate*)—current available on each power rail is outlined in the table below. These circuits are protected with self resetting fuses. These circuits can also be toggled on and off through ROS.

User Power Rail	Maximum (Aggregate) Current
12 V Rail A	5 A
12 V Rail B	5 A
24 V	5 A
VBAT (24 - 29 V, unregulated)	5 A

 DANGER

Do not exceed the specified maximum current limit of each power rail. Exceeding the maximum current may damage to your payload.

2.3.1.1 User Power Pinout

The pinout for each of the 8 connectors—(*USER 1 through USER 8*)—is illustrated below.

USER connector pin	Function
1	12 V (<i>Rail B</i>)
2	12 V (<i>Rail A</i>)
3	24 V
4	VBAT (<i>24 - 29 V, unregulated</i>)
5	GND
6	GND
7	GND
8	GND

► Making User Power Cables

2.3.1.2 Software Control of User Power Rails

User power can be enabled or disabled through ROS by publishing to

`/a300_####/platform/mcu/cmd_user_power` ROS topic.

- To disable user power, run:

```
ros2 topic pub --once /a300_#####/platform/mcu/cmd_user_power
std_msgs/msg/Bool '{data: false}'
```

- To enable user power, run:

```
ros2 topic pub --once /a300_#####/platform/mcu/cmd_user_power
std_msgs/msg/Bool '{data: true}'
```

NOTE

User Power is enabled by default shortly after the robot first powers up, regardless of its state when the robot was powered off.

2.3.2 Expansion Power (*EXP 1 and EXP 2*)

There are 2 Expansion Power connectors on the System Interface circuit board. The Expansion Power connectors can power small manipulators, DC-AC inverters, and DC-DC supplies.

These connectors supply VBAT voltage, with a combined current of up to 20 A. The connectors also include soft-start circuitry which limits inrush current to 25 A. This soft-start circuit reduces the chance of a high capacitance load causing the 20 A fuse to pop.

REPLACEMENT FUSE

Use a [Littelfuse 0297020.U](#) as the replacement.

2.3.2.1 Expansion Power Pinout

The pinout for each of the 2 connectors—(*EXP 1 and EXP 2*)—is illustrated below.

Expansion Power connector pin	Function
1	VBAT (24 - 29 V, unregulated)

Expansion Power connector pin	Function
2	GND

2.3.2.2 Software Control of Expansion Power Rails

The Expansion Power connectors are connected to the same control system as the User Power. Refer to the [Software Control of User Power Rails](#) section for details.

2.3.3 High Power Expansion Kit (*Large VBAT*)

NOTE

The optional High Power Expansion Kit can be purchased at the time of your original Husky order —(*Clearpath item 032001*). We do not allow field installation of the kit.

TIP

We suggest you purchase a 80 Ah or 120 Ah Husky if you plan to operate on hills steeper than 5°, and have a payload of more than 50kg. 40 Ah battery configuration may not provide enough power in the worst case. Contact [Support](#) for more details.

Some high power payloads, such as manipulators, require more power than can be provided through the User Power connectors. The optional High Power Expansion Kit provides VBAT voltage at 40 A. It also includes 2 contactors for enabling and disabling power to the payload.

► [Making A Large VBAT Cable](#)

REPLACEMENT FUSE

Use a [Littelfuse 142.5631.5402](#) as the replacement.

2.3.3.1 On/Off Control via Software

The High Power Expansion Kit output is controlled through software by publishing to the ROS topic `/a300_####/platform/mcu/cmd_high_power`, similar to [Software Control of User Power Rails](#).

NOTE

The High Power Expansion Kit output is always disabled when the system powers up. You must enable the output, via the ROS topic above, each time after the system powers up.

2.4 AUX Inputs

The System Interface circuit board has 3 auxiliary inputs—(*AUX1 IN*, *AUX2 IN*, *AUX3 IN*)—that allow external payloads to provide inputs to the robot. Each of these auxiliary inputs can be driven by one of the following:

- a 12 V input
- a 24 V input
- a push button or switch

The state of the input can be read from the ROS topic `/a300_####/platform/mcu/status/aux_inputs`. The AUX inputs are active low and are mapped to the ROS message bits as follows:

- AUX1 IN: bit 0
- AUX2 IN: bit 1
- AUX3 IN: bit 2

AUX Input	Payload State	<code>/a300_####/platform/mcu/status/aux_inputs</code> Value
AUX1 IN	24 V on OR 12 V on OR button pressed 24 V off AND 12 V off AND button not pressed	bit 0 reads as '0' bit 0 reads as '1'

AUX Input	Payload State	/a300_####/platform/mcu/status/aux_inputs Value
AUX2 IN	24 V on OR 12 V on OR button pressed 24 V off AND 12 V off AND button not pressed	bit 1 reads as '0' bit 1 reads as '1'
AUX3 IN	24 V on OR 12 V on OR button pressed 24 V off AND 12 V off AND button not pressed	bit 2 reads as '0' bit 2 reads as '1'

2.4.1 AUX Input Pinout

The pinout for each of the connectors—(AUX1 IN, AUX2 IN, AUX3 IN)—is illustrated below.

AUX IN Connector's Pin	Function
1	24 V In
2	12 V In
3	Button/Switch In, (3.3 V)
4	24 V Return
5	12 V Return
6	GND

▶ AUX Input Example #1

▶ AUX Input Example #2

2.5 AUX Outputs

The System Interface circuit board includes 3 outputs—(*AUX1 OUT*, *AUX2 OUT*, *AUX3 OUT*)—that can be controlled by publishing to the `/a300_####/platform/mcu/cmd_aux_outputs` ROS topic. To enable the AUX outputs, publish a single message with the appropriate bits set to 1 to turn the output on, and 0 to turn it off. The state does not persist so new commands will need to keep the bit enabled as well. The AUX outputs are mapped to the ROS message bits as follows:

- AUX1 OUT: bit 0
- AUX2 OUT: bit 1
- AUX3 OUT: bit 2

2.5.1 AUX Output Pinout

Each of the AUX outputs is an 8 position connector with pinout as outlined below. The output value is available in several modes:

- MOSFET outputs, (2X)
- Normally open relay
- Normally closed relay

AUX OUT Connector Pin	Name	Function	Output when corresponding <code>cmd_aux_outputs</code> bit is '0'	Output when corresponding <code>cmd_aux_outputs</code> bit is '1'
1	AUX_24V	24 V output, fused at 3 A	24 V	24 V
2	GND	GND	GND	GND
3	AUX_RELAY_NC	Normally closed relay pin	Connected to AUX_RELAY_COMMON	Floating

AUX OUT Connector Pin	Name	Function	Output when corresponding <code>cmd_aux_outputs</code> bit is '0'	Output when corresponding <code>cmd_aux_outputs</code> bit is '1'
4	AUX_MOSFET2	MOSFET #2 DRAIN	GND	Floating
5	GND	GND	GND	GND
6	AUX_RELAY_COMMON	Input that gets connected to AUX_RELAY_NC or AUX_RELAY_NO	Connected to AUX_RELAY_NC	Connected to AUX_RELAY_NO
7	AUX_RELAY_NO	Normally open relay pin	Floating	Connected to AUX_RELAY_NC
8	AUX_MOSFET1	MOSFET #1 DRAIN	GND	Floating

▶ AUX Output Examples

2.6 General Purpose Inputs and Outputs (GPIO)

The System Interface circuit board includes 4 general purpose inputs, and 4 general purpose outputs. All GPIOs are at the 3.3 V logic level. General purpose inputs are readable from the `/a300_####/platform/mcu/status/gp_inputs` ROS topic. Publishing to the `/a300_####/platform/mcu/cmd_gp_outputs` ROS topic controls the general purpose outputs.

! INFO

All the inputs and outputs are exposed through a single connector.

2.6.1 General Purpose Input Pinout

GPIO connector pin	Function	gp_inputs value when driven at 0 V	gp_inputs value when driven at 3.3 V
5	GP In 1	Bit 0 is '0'	Bit 0 is '1'
6	GP In 2	Bit 1 is '0'	Bit 1 is '1'
7	GP In 3	Bit 2 is '0'	Bit 2 is '1'
8	GP In 4	Bit 3 is '0'	Bit 3 is '1'
13	GND	N/A	N/A
14	GND	N/A	N/A
15	GND	N/A	N/A
16	GND	N/A	N/A

2.6.2 General Purpose Output Pinout

GPIO connector pin	Function	Corresponding bit in cmd_gp_outputs	Output value when set to '0'	Output value when set to '1'
1	GP Out 1	0	0 V	3.3 V
2	GP Out 2	1	0 V	3.3 V
3	GP Out 3	2	0 V	3.3 V
4	GP Out 4	3	0 V	3.3 V
9	GND	N/A	GND	GND

GPIO connector pin	Function	Corresponding bit in <code>cmd_gp_outputs</code>	Output value when set to '0'	Output value when set to '1'
10	GND	N/A	GND	GND
11	GND	N/A	GND	GND
12	GND	N/A	GND	GND

2.7 Fan Expansion

The System Interface circuit board has support for 4 additional 12 V brushless 4-wire PWM fans. Ports FAN 1, FAN 2, FAN 3, and FAN 4 are already used by the base Husky A300. Ports FAN 5, FAN 6, FAN 7, and FAN 8 are available for your fan expansions.

❗ EXAMPLES OF SUPPORTED FANS

- [CFM-8025BG-170-517-22](#)
- [9GA0812P4J001](#).

The pinout for each fan is:

Pin	Function
1	12 V
2	GND
3	Tachometer (<i>output from fan</i>)
4	PWM (<i>input to fan, 3.3 V</i>)

The speed of each fan can be controlled by publishing to the

`/a300_####/platform/mcu/cmd_fans` ROS topic.

The speed of each fan can be monitored by reading the `/a300_####/platform/mcu/status/fans` ROS topic.

In both cases, the fan value is in the range from 0 (*not spinning*), to 255 (*max speed*).

! INFO

Fans require spool up time on startup. Setting a fan to a very low value may result in the fan not rotating. You should experiment to find minimum PWM value that causes your fan to spin.

2.8 Wireless Emergency Stop

A wireless dual-channel emergency stop can be added to Husky. This allows you to invoke an emergency stop remotely—(*either through a button press or when the remote unit is out of range*). This is in addition to the Husky's built-in Emergency Stop Buttons. Refer to the User Manual for details on how the Wireless Emergency Stop can be bypassed in certain debugging scenarios.

The Wireless Emergency Stop can be connected to Husky on the System Interface circuit board at the **W ES** connector, based on the pinout below.

W ES Pin	Function / Value
1	24 V, (<i>fused at 3 A</i>)
2	CAN High: combined with CAN Low, connected to the CAN2 pins on the MCU; see additional notes below
3	Reset In: Pull to GND for at least 100 ms to restart the robot after all emergency stops have been cleared; equivalent to pressing the Restart Button on Husky
4	Channel 1 Out, (24 V): normally connected to Channel 1 In; disconnect to trigger an emergency stop
5	Channel 1 In

W ES Pin	Function / Value
6	GND
7	CAN Low
8	GND
9	Channel 2 Out, (24 V): normally connected to Channel 2 In; disconnect to trigger an emergency stop
10	Channel 2 In

2.8.1 Receiver Installation

In addition to the electrical integration noted above, the wireless receiver will normally require mechanical mounting inside the Husky, along with external mounting of an antenna. Refer to Wireless Stop receivers and Cable Passthrough for additional details.

2.9 CANbus Connection

Husky has up to 3 standard CAN networks:

Network name	Data Rate	Purpose	Devices
vcan0	1000 kBit/s	Motor Control	Primary Computer, MCU, 4 Motor Controllers
vcan1	250 kBit/s	Batteries	Primary Computer, MCU, Batteries (<i>up to 6 batteries</i>), Wireless Emergency Stop
usbcan0	250 kBit/s	Optional Wireless Charger	Primary Computer, Wireless Charger

! INFO

We suggest you add custom CAN devices to a new network, using a USB-CAN adapter such as a PEAK System PCAN-USB. You would need to name the new network something different than the 3 networks listed above. We configure these networks using SocketCAN.

It is possible to add new devices to the Husky's existing networks, but this requires significant testing time.

2.10 Additional Emergency Stop Devices

The base Husky A300 includes 2 Emergency Stop Buttons, which are connected to the System Interface circuit board. The System Interface circuit board also includes connector breakouts for 6 more Emergency Stop devices, labelled ES3 through ES8. The standard Husky A300 has a bypass jumper inserted into each of these unused Emergency Stop breakouts.

You can use these Emergency Stop breakouts to add more Emergency Stop devices, such as:

- Emergency Stop buttons
- Safety lidar
- Tape switches
- Limit switches
- Relays or similar PLC components
- Manipulator control cabinets

PINOUT FOR ES3 - ES8

ES Pin	Function / Value
1	Channel 1 Out, (24 V): normally connected to Channel 1 In; disconnect to trigger an emergency stop
2	Channel 1 In

ES Pin	Function / Value
3	Channel 2 Out (24 V): normally connected to Channel 2 In; disconnect to trigger an emergency stop
4	Channel 2 In

2.11 System Power

These 4 connectors collectively provide up to 400 W of power at 12 V for powering computers and network equipment.

2.11.1 SYS1 And SYS2 Pinouts For Computers

Pin	Function / Value
1	12 V
2	12 V
3	GND
4	GND

INFO

SYS1 is always being used by the robot's Primary Computer.

2.11.2 SYS3 Pinout For A Graphics Card

Pin	Function / Value
1	12 V

Pin	Function / Value
2	12 V
3	12 V
4	GND
5	GND
6	GND

2.11.3 SYS4 System Power For A Network Switch

This provides 60 W of power at 12 V.

Pin	Function / Value
1	12 V
2	GND

INFO

SYS4 is always being used power the Husky's network switch.

3. Software Integration

ROS has a large ecosystem of sensor drivers, some of which include pre-made URDF descriptions and even simulation configurations. Refer to [Sensors supported by ROS](#).

For the best experience, consider purchasing supported accessories from Clearpath Robotics for your robot, which will include simulation, visualization, and driver support.

Refer to the following for more details:

- [Computers](#)
- [Sensors](#)
- [Manipulators](#)
- [Accessories](#)

4. Support

Clearpath is committed to your success. Please get in touch with us and we will do our best to get you rolling again quickly: support@clearpathrobotics.com.

To get in touch with a salesperson regarding Clearpath Robotics products, please email research-sales@clearpathrobotics.com.

If you have an issue that is specifically about ROS and is something which may be of interest to the broader community, consider asking it on [Robotics Stack Exchange](#). If you do not get a satisfactory response, please ping us and include a link to your question as posted there. If appropriate, we will answer on Robotics Stack Exchange for the benefit of the community.

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